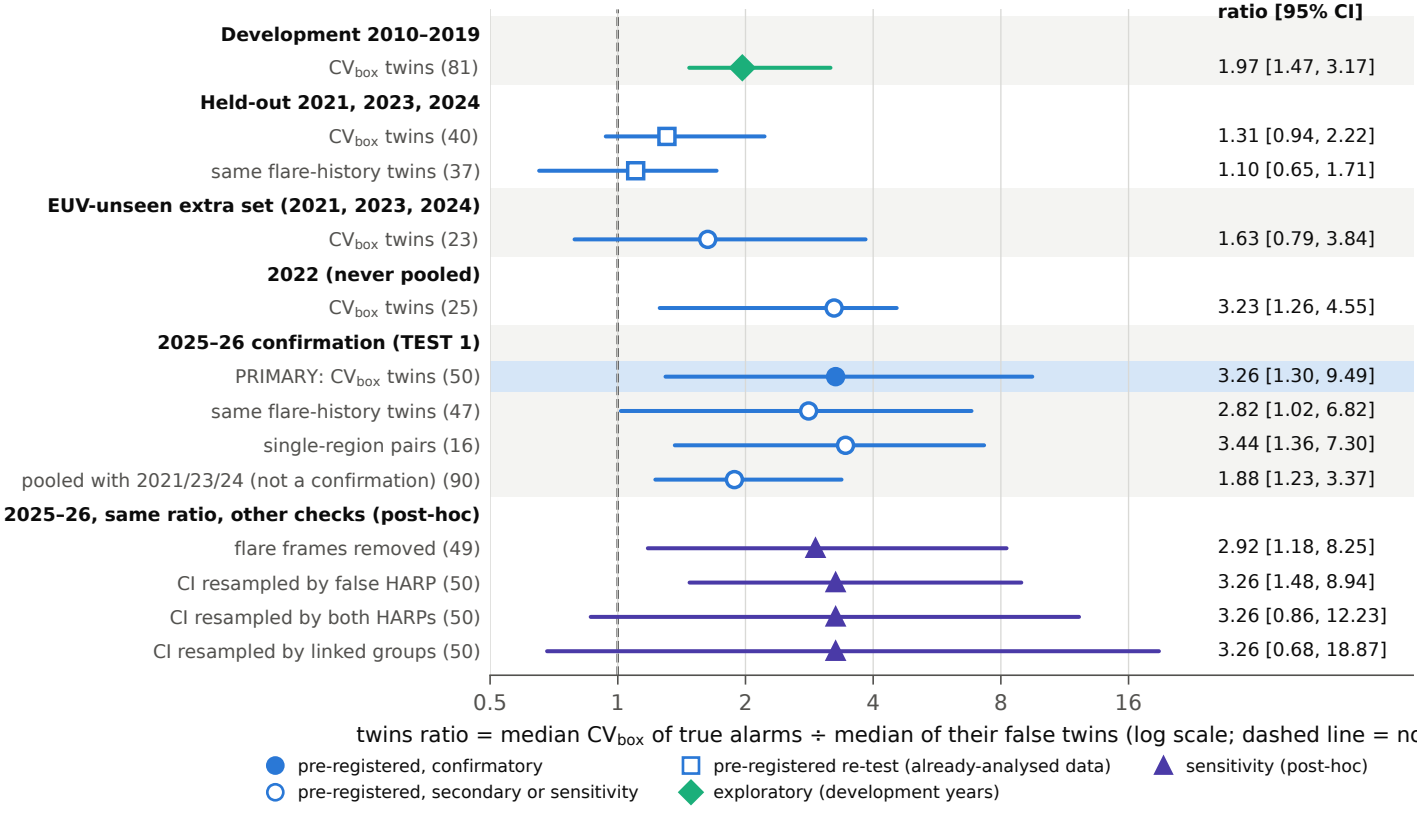
ratio [95% CI]
Development 2010–2019
CVbox twins (81)
1.97 [1.47, 3.17]
Held-out 2021, 2023, 2024
CVbox twins (40)
1.31 [0.94, 2.22]
same flare-history twins (37)
1.10 [0.65, 1.71]
EUV-unseen extra set (2021, 2023, 2024)
CVbox twins (23)
1.63 [0.79, 3.84]
2022 (never pooled)
CVbox twins (25)
3.23 [1.26, 4.55]
2025–26 confirmation (TEST 1)
PRIMARY: CVbox twins (50)
3.26 [1.30, 9.49]
same flare-history twins (47)
2.82 [1.02, 6.82]
single-region pairs (16)
3.44 [1.36, 7.30]
pooled with 2021/23/24 (not a confirmation) (90)
1.88 [1.23, 3.37]
2025–26, same ratio, other checks (post-hoc)
flare frames removed (49)
2.92 [1.18, 8.25]
CI resampled by false HARP (50)
3.26 [1.48, 8.94]
CI resampled by both HARPs (50)
3.26 [0.86, 12.23]
CI resampled by linked groups (50)
3.26 [0.68, 18.87]
0.5
1
2
4
8
16
twins ratio = median CVbox of true alarms ÷ median of their false twins (log scale; dashed line = no
pre-registered, confirmatory
pre-registered, secondary or sensitivity
pre-registered re-test (already-analysed data)
exploratory (development years)
sensitivity (post-hoc)